DOMAIN 5: PORTFOLIO ENGINE, QUANTITATIVE OPTIMIZATION & TRACKING

13 Core Tables • SLSQP Integer Lots, Immutable Versions, Mathematical Invariant Proofs & Explainability Evidence

13 TABLES

investor_portfolios — Multi-Tenant Portfolio Root

PK	**portfolio_id**	UUID
FK	**user_id**	UUID
	portfolio_name	VARCHAR
	portfolio_status	VARCHAR
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

portfolio_generation_requests — Agent Mandate

PK	**generation_request_id**	UUID
FK	**portfolio_id**	UUID
FK	**user_id**	UUID
FK	**goal_id**	UUID
FK	**risk_assessment_id**	UUID
	request_type	VARCHAR
	request_status	VARCHAR
	user_prompt	VARCHAR
	requested_amount	NUMERIC
	currency_code	VARCHAR
	investment_horizon_months	INTEGER
	normalized_inputs	JSONB
	investor_snapshot	JSONB
	market_context_as_of	TIMESTAMP
	started_at	TIMESTAMP
	completed_at	TIMESTAMP
	error_code	VARCHAR
	error_summary	VARCHAR
	created_at	TIMESTAMP

portfolio_versions — Immutable Version State

PK	**portfolio_version_id**	UUID
FK	**portfolio_id**	UUID
FK	**generation_request_id**	UUID
FK	**strategy_version_id**	UUID
	version_number	INTEGER
	version_status	VARCHAR
	investment_amount	NUMERIC
	currency_code	VARCHAR
	investment_horizon_months	INTEGER
	risk_category	VARCHAR
	expected_return_pct	NUMERIC
	expected_volatility_pct	NUMERIC
	diversification_score	NUMERIC
	market_context_as_of	TIMESTAMP
	generation_context	JSONB
	generated_at	TIMESTAMP
	created_at	TIMESTAMP

portfolio_security_decisions — Integer Allocations (Q_i)

PK	**decision_id**	UUID
FK	**portfolio_version_id**	UUID
FK	**security_id**	UUID
FK	**listing_id**	UUID
	decision_status	VARCHAR
	target_weight_pct	NUMERIC
	decision_reason	VARCHAR
	created_at	TIMESTAMP

strategy_definitions — SLSQP Optimizer Config

PK	**strategy_id**	UUID
	strategy_code	VARCHAR
	strategy_name	VARCHAR
	description	VARCHAR
	is_active	BOOLEAN
	created_at	TIMESTAMP

strategy_versions — Fiduciary Strategy Specs

PK	**strategy_version_id**	UUID
FK	**strategy_id**	UUID
	version_label	VARCHAR
	optimizer_name	VARCHAR
	implementation_revision	VARCHAR
	configuration	JSONB
	constraint_set_version	VARCHAR
	created_at	TIMESTAMP

portfolio_version_constraints — Hard Bounds & Caps

PK	**portfolio_constraint_id**	UUID
FK	**portfolio_version_id**	UUID
	constraint_code	VARCHAR
	operator	VARCHAR
	comparison_value	NUMERIC
	upper_value	NUMERIC
	unit_code	VARCHAR
	is_mandatory	BOOLEAN
	applied	BOOLEAN
	description	VARCHAR
	created_at	TIMESTAMP

portfolio_scenarios — Stress-Test Projections

PK	**portfolio_scenario_id**	UUID
FK	**portfolio_version_id**	UUID
	scenario_code	VARCHAR
	scenario_name	VARCHAR
	scenario_type	VARCHAR
	assumptions	JSONB
	estimated_portfolio_value	NUMERIC
	estimated_change_amount	NUMERIC
	estimated_change_pct	NUMERIC
	result_summary	VARCHAR
	calculation_version	VARCHAR
	created_at	TIMESTAMP

recommendation_explanations — Level-1 Thesis

PK	**explanation_id**	UUID
FK	**portfolio_version_id**	UUID
FK	**decision_id**	UUID
	explanation_type	VARCHAR
	title	VARCHAR
	explanation_text	VARCHAR
	display_order	INTEGER
	created_at	TIMESTAMP

explanation_evidence — Thesis to Data Citation

PK, FK	**explanation_id**	UUID
PK, FK	**evidence_item_id**	UUID
	relationship_label	VARCHAR

evidence_items — Level-2 Proof Registry

PK	**evidence_item_id**	UUID
FK	**data_source_id**	UUID
FK	**security_id**	UUID
FK	**market_price_id**	UUID
FK	**fundamental_metric_id**	UUID
FK	**economic_observation_id**	UUID
FK	**news_article_id**	UUID
	evidence_type	VARCHAR
	title	VARCHAR
	evidence_summary	VARCHAR
	source_url	VARCHAR
	source_record_reference	VARCHAR
	published_at	TIMESTAMP
	observed_at	TIMESTAMP
	cited_value_snapshot	JSONB
	created_at	TIMESTAMP

portfolio_validation_runs — Deterministic Math Audit

PK	**validation_run_id**	UUID
FK	**portfolio_version_id**	UUID
	rule_set_version	VARCHAR
	calculation_engine_version	VARCHAR
	validation_status	VARCHAR
	started_at	TIMESTAMP
	completed_at	TIMESTAMP
	summary	JSONB

portfolio_validation_results — Post-Rounding Cap Checks

PK	**validation_result_id**	UUID
FK	**validation_run_id**	UUID
	rule_code	VARCHAR
	result_status	VARCHAR
	severity	VARCHAR
FK	**security_id**	UUID
	actual_value	NUMERIC
	threshold_value	NUMERIC
	result_message	VARCHAR
	details	JSONB
	created_at	TIMESTAMP